06 – Computer architecture & history of computing

A software needs a hardware to run on…

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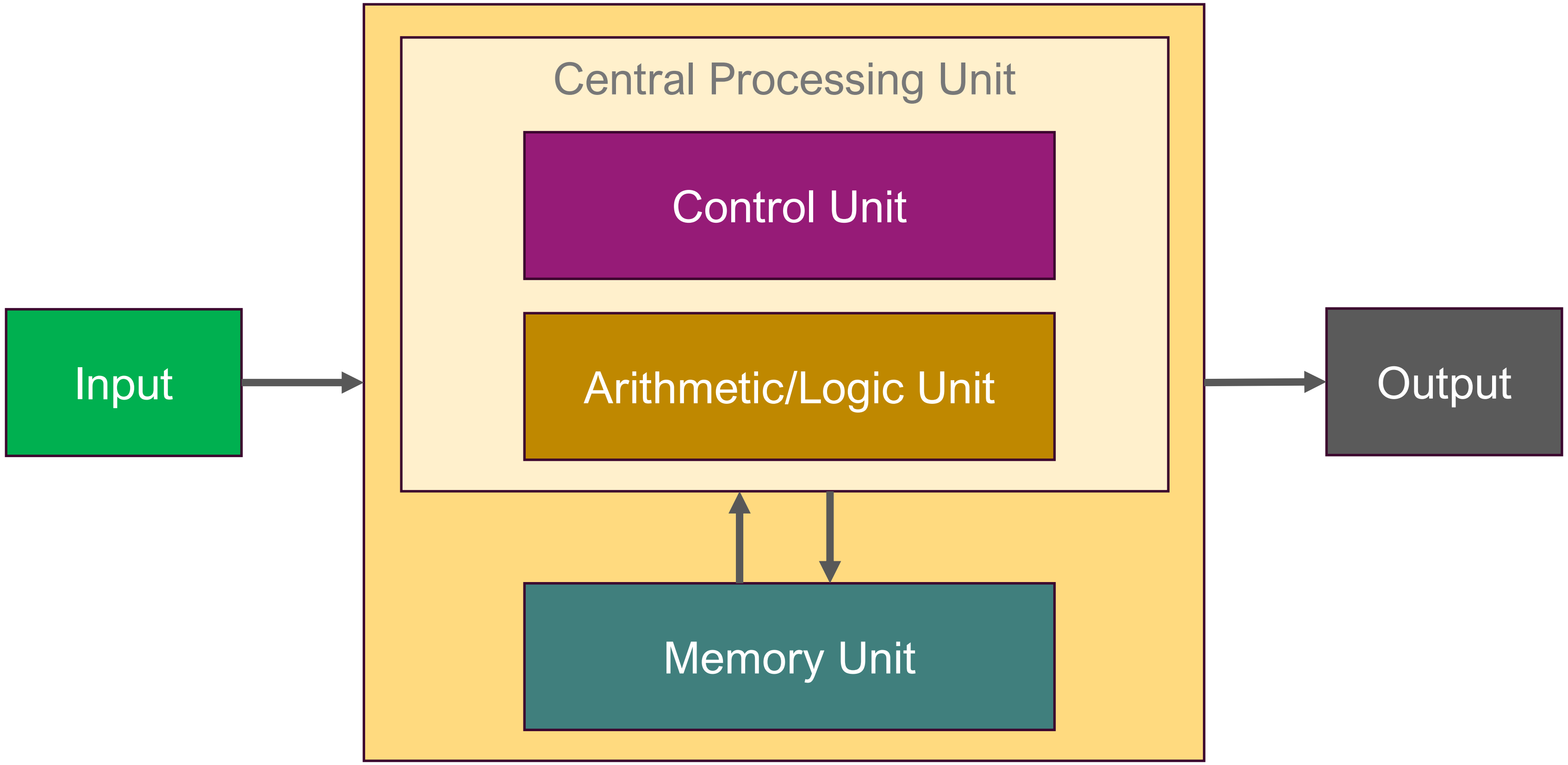
OUTLINE

1) Computer Architecture
2) History of Computing
3) Operating Systems

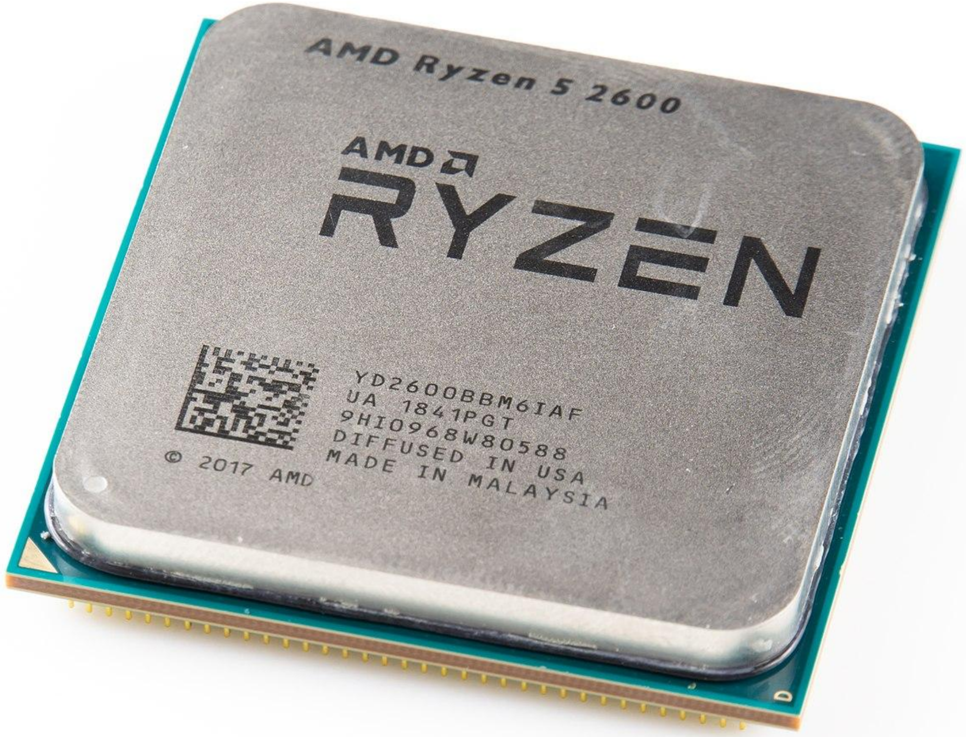
1) COMPUTER ARCHITECTURE

- A computer is built on the Von Neumann architecture defined by John Von Neumann around 1945, https://en.wikipedia.org/wiki/Von_Neumann_architecture



1) COMPUTER ARCHITECTURE

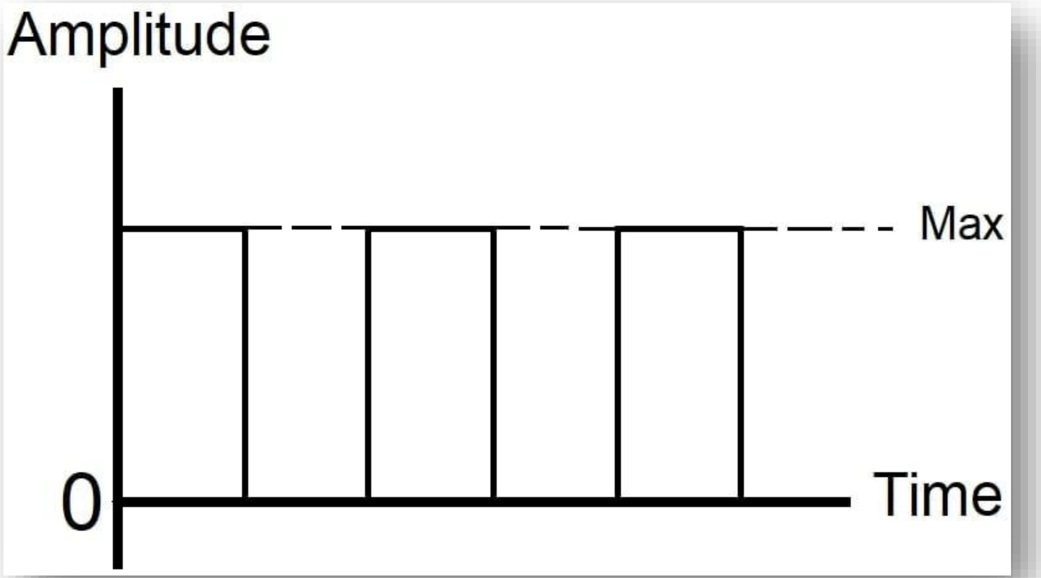
- Microprocessor:
  - The microprocessor is a multipurpose, clock-driven, register-based, digital integrated circuit that accepts binary data as input, processes it according to instructions stored in its memory, and provides results (also in binary form) as output. Microprocessors contain both combinational logic and sequential digital logic and, operate on numbers and symbols represented in the binary number system.



Source: https://en.wikipedia.org/wiki/Microprocessor

1) COMPUTER ARCHITECTURE

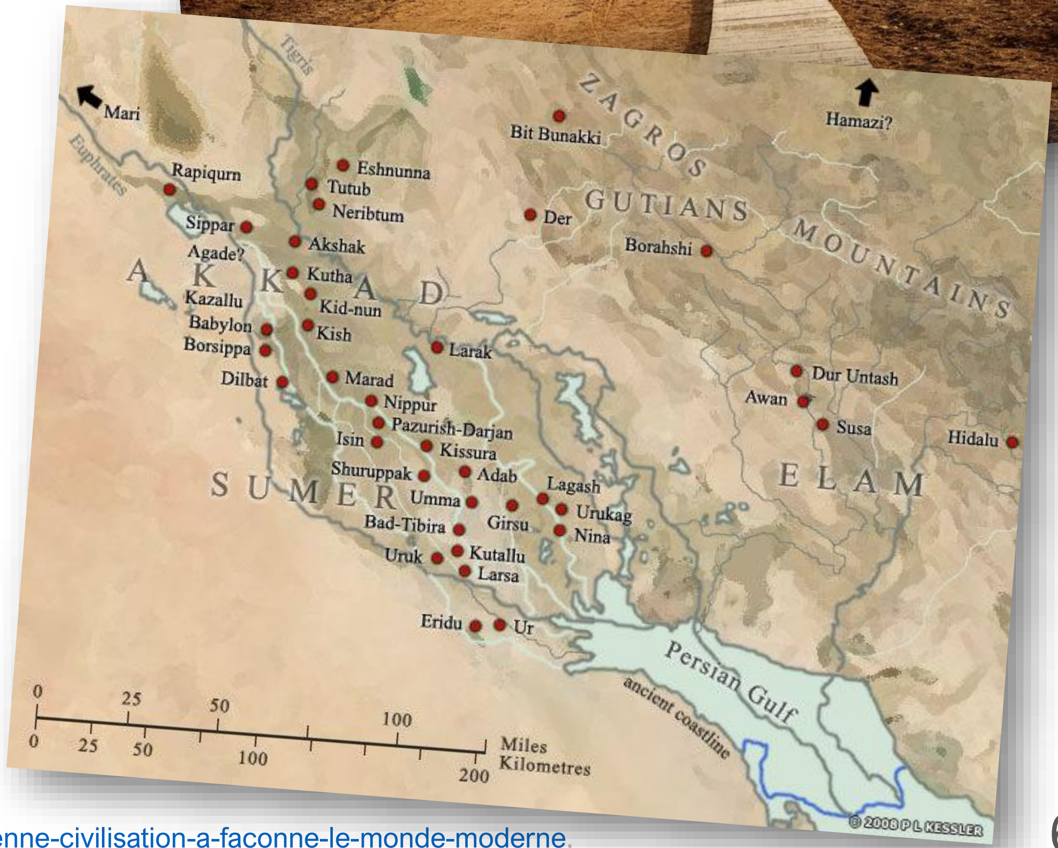
- Binary number system:
  - Two states: 0/1, False/True, Off/On, Low/High, Open/Closed
- 00111010100101010101001010101
  01010101010101010101010101010
  10101111010101110101010111010
  10101101011010101110110101011
  010101010110
- Easy to do in electronics:
  - Current in a cable or not



Source: https://eepower.com/technical-articles/digital-electronics-basics-digital-binary-operation

2) HISTORY OF COMPUTING

- Since the Antiquity (Ancient Ages), human beings have always used computation to solve everyday life problems
- The development of Mathematics means the development of Techniques and the improvement of the Quality of life
- Sumer was the first city to use balls and clay tokens to process information around 10 000 BC, Sumer also invented writing around 3 200 BC



Source: https://www.nationalgeographic.fr/histoire/culture-generale-ancetres-moyen-orient-cette-ancienne-civilisation-a-faconne-le-monde-moderne, https://www.worldhistory.org/image/1352/map-of-sumer/

2) HISTORY OF COMPUTING

- Egypt used balls and clay tokens for surveying during Nile floods around 3 000 BC

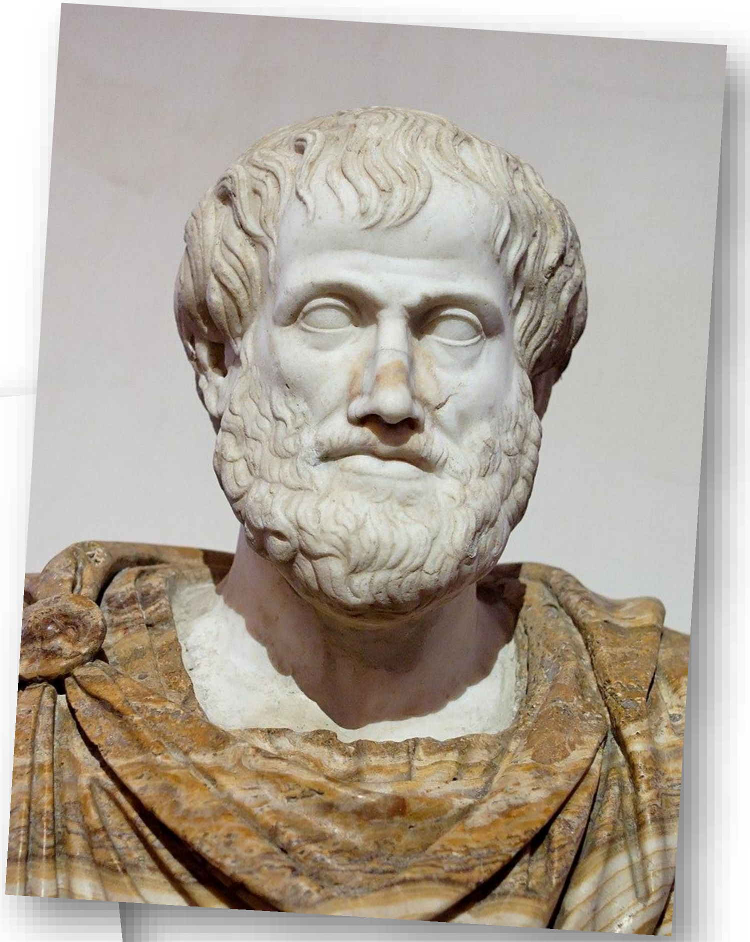
- The abacus appears in China around 3 000 BC

Sources: https://en.wikipedia.org/wiki/Egyptian_pyramids#/media/File:All_Gizah_Pyramids.jpg, https://www.amazon.com/Forti-Wooden-Abacus-Chinese-Calculator/dp/B006RYZ3YQ

2) HISTORY OF COMPUTING

- In Greece, Aristotle defines the logic reasoning around 330 BC.

$$
\begin{array}{l}
p \\
\underline{p \to q} \\
\therefore q
\end{array}
$$



Source: https://en.wikipedia.org/wiki/Logic

2) HISTORY OF COMPUTING

- The roman abacus is the 100 AD version of the chinese one
- The Persian Al-Khwarizmi, linear and quadratic equations solutions, near 820 AD: Algorithm
- Fibonacci and its sequence from 1202 AD in "Liber Abaci":

1,1,2,3,5,8,13,21,34,55,89,144,233,377, 610,987,…

Sources: https://www.worldhistory.org/image/2147/roman-abacus/, https://en.wikipedia.org/wiki/Al-Khwarizmi, https://www.thoughtco.com/leonardo-pisano-fibonacci-biography-2312397

2) HISTORY OF COMPUTING

- In South America, the Quipu memorizes information with knots on strings (from 2 600 BC to 17th century)
- Rediscovery of the golden ratio by Luca Pacioli, in 1509 AD:

$$\varphi^2 = \varphi + 1$$

$$\varphi = \frac{1+\sqrt{5}}{2}$$

Sources: https://en.wikipedia.org/wiki/Quipu, https://en.wikipedia.org/wiki/Golden_ratio

2) HISTORY OF COMPUTING

- In 1614, the Scottish John Napier invents the logarithm:

$$\log(ab) = \log a + \log b$$

- In 1623, Wilhelm Schickard invents the first mechanical calculator with two operations: addition and subtraction

Source: https://en.wikipedia.org/wiki/John_Napier, https://en.wikipedia.org/wiki/Wilhelm_Schickard

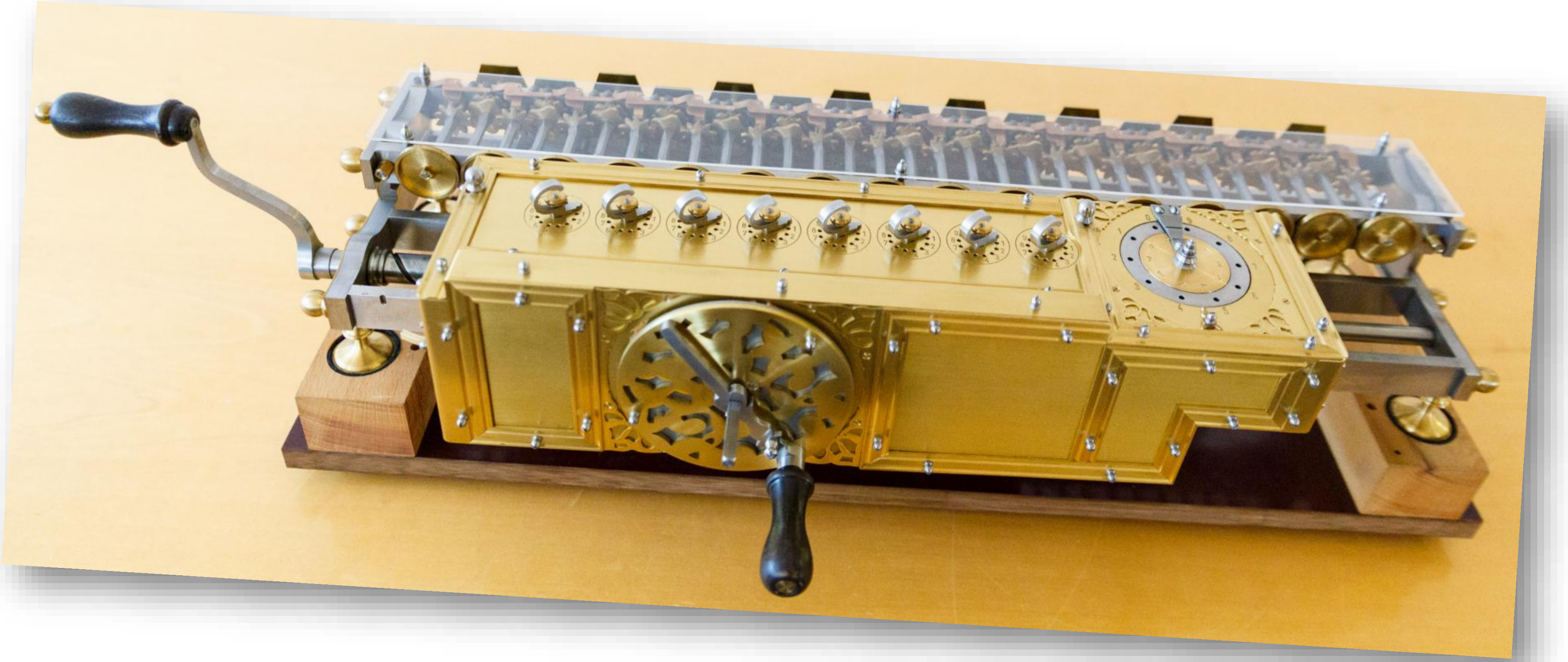
2) HISTORY OF COMPUTING

- In 1642, Blaise Pascal creates the ”Pascaline”, a mechanical calculator
- Pascal was not even 19 years old!
- To help his father to get tax paid…

Source: https://en.wikipedia.org/wiki/Blaise_Pascal

2) HISTORY OF COMPUTING

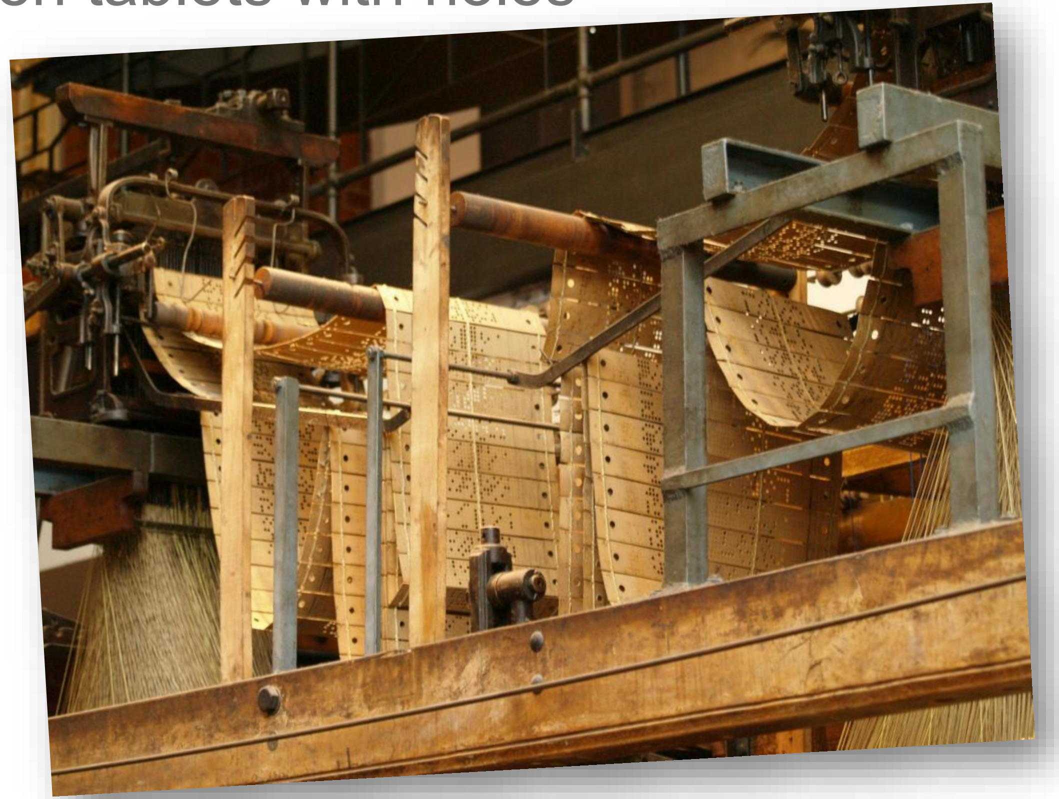
- In 1673, Gotfried Von Leibniz creates a mechanical calculator able of all four operations: addition, subtraction, multiplication and division



Source: https://commons.wikimedia.org/wiki/File:Rechenmaschine_von_Leibniz_%28Nachbau%29_04.jpg

2) HISTORY OF COMPUTING

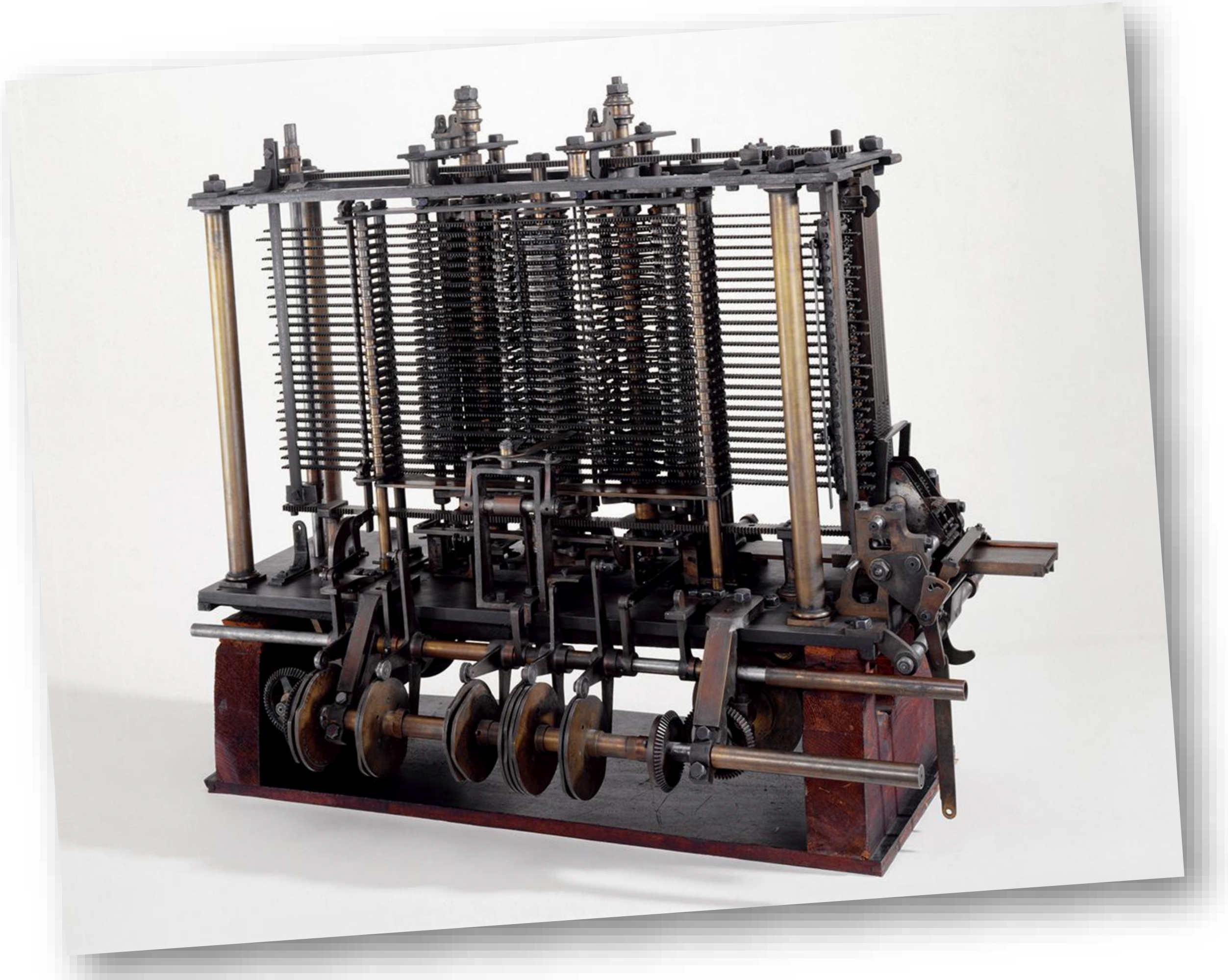
- In 1806, the Jacquard machine (to make fabric) is a mechanical machine **programmable** with wooden tablets with holes



Source: https://en.wikipedia.org/wiki/Jacquard_machine

2) HISTORY OF COMPUTING

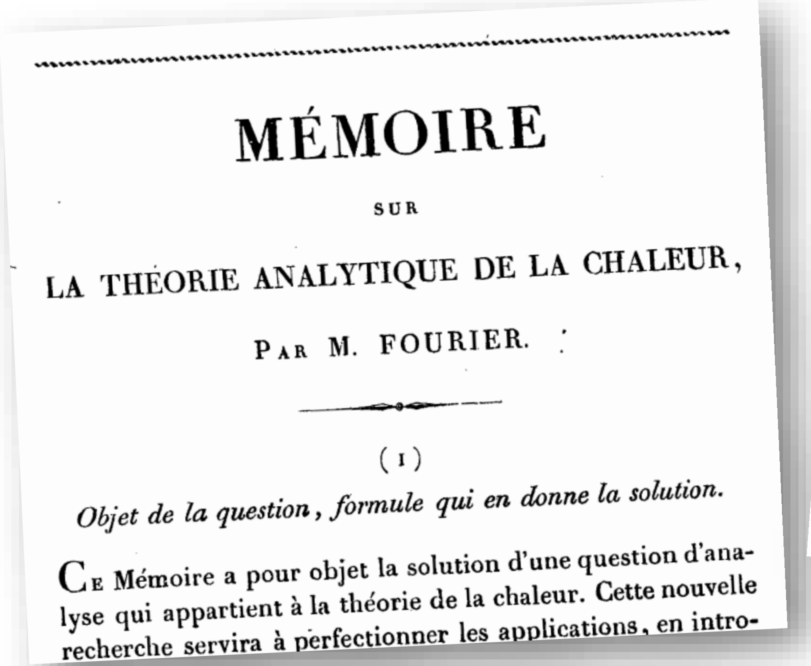
- In 1820, Charles Babbage invents the analytical engine, a **mechanical programmable computer**
- 25 000 elements but it didn't work at this time, it was built in 1991 by a British university: it works!
- Ada Lovelace in 1842 proposed a program to compute Bernouilli numbers to Charles Babbage: she is the first programmer in History!



Source: https://en.wikipedia.org/wiki/Analytical_engine

2) HISTORY OF COMPUTING

- In 1822, in ”Théorie analytique de la chaleur”, Joseph Fourier proposes the Fourier Transform that is still used as a tool in signal processing



MÉMOIRE

SUR

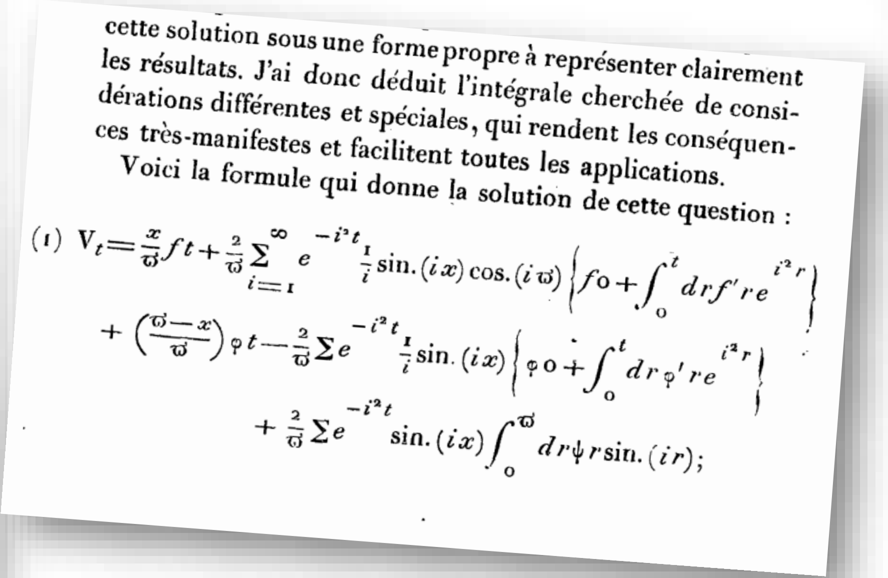
LA THÉORIE ANALYTIQUE DE LA CHALEUR,

PAR M. FOURIER.

(1)

Objet de la question, formule qui en donne la solution.

CE Mémoire a pour objet la solution d'une question d'analyse qui appartient à la théorie de la chaleur. Cette nouvelle recherche servira à perfectionner les applications, en intro-



cette solution sous une forme propre à représenter clairement les résultats. J'ai donc déduit l'intégrale cherchée de considérations différentes et spéciales, qui rendent les conséquences très-manifestes et facilitent toutes les applications.

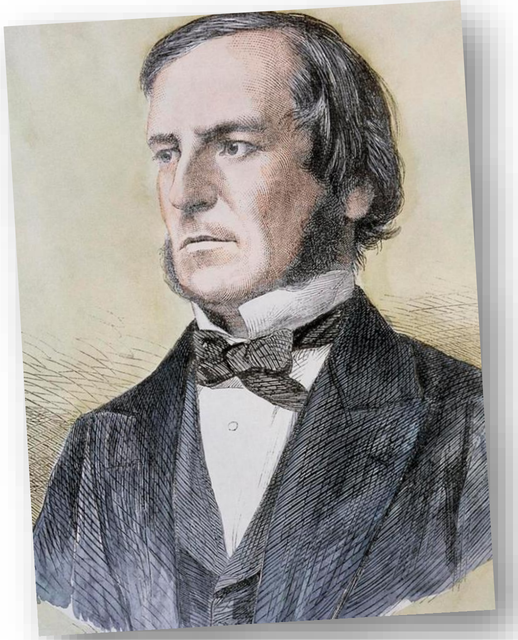
Voici la formule qui donne la solution de cette question :

$$(1)\quad V_t = \frac{x}{\varpi} f t + \frac{2}{\varpi}\sum_{i=1}^{\infty} e^{-i^2 t}\frac{1}{i}\sin.(ix)\cos.(i\varpi)\left\{fo + \int_0^t dr f' r e^{i^2 r}\right\}$$
$$+ \left(\frac{\varpi - x}{\varpi}\right)\varphi t - \frac{2}{\varpi}\Sigma e^{-i^2 t}\frac{1}{i}\sin.(ix)\left\{\varphi o + \int_0^t dr \varphi' r e^{i^2 r}\right\}$$
$$+ \frac{2}{\varpi}\Sigma e^{-i^2 t}\sin.(ix)\int_0^{\varpi} dr \psi r \sin.(ir);$$

Source: https://en.wikipedia.org/wiki/Joseph_Fourier, https://en.wikipedia.org/wiki/Fourier_series

2) HISTORY OF COMPUTING

- In 1854, George Boole introduces the Boolean algebra using logic and the binary system: 0/1

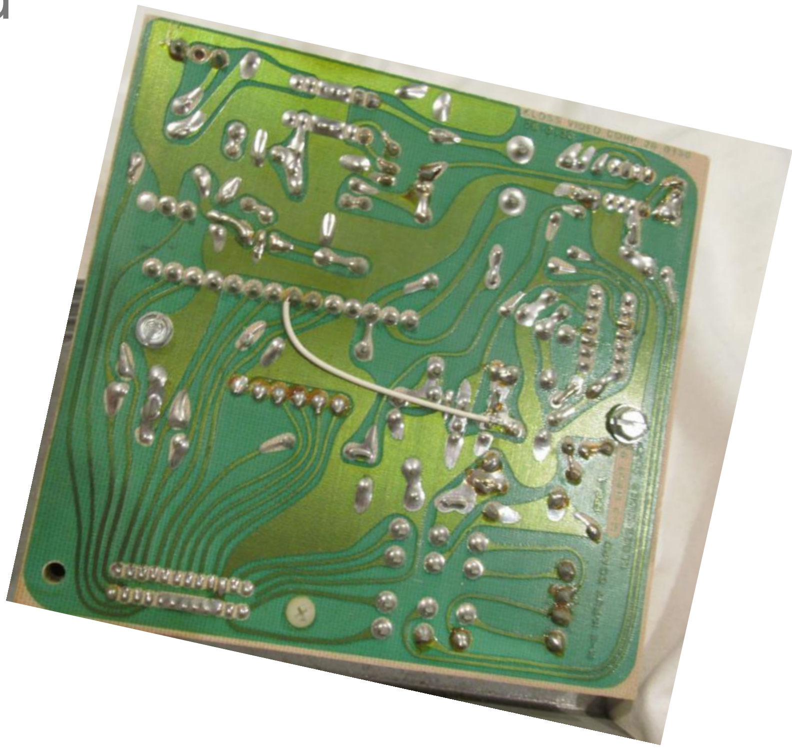


Function	x	0	0	1	1
	y	0	1	0	1
Constant 0	0	0	0	0	0
And	$x \cdot y$	0	0	0	1
x And Not y	$x \cdot \bar{y}$	0	0	1	0
x	x	0	0	1	1
Not x And y	$\bar{x} \cdot y$	0	1	0	0
y	y	0	1	0	1
Xor	$x \cdot \bar{y} + \bar{x} \cdot y$	0	1	1	0
Or	$x + y$	0	1	1	1
Nor	$\overline{x + y}$	1	0	0	0
Equivalence	$x \cdot y + \bar{x} \cdot \bar{y}$	1	0	0	1
Not y	$\bar{y}$	1	0	1	0
If y then x	$x + \bar{y}$	1	0	1	1
Not x	$\bar{x}$	1	1	0	0
If x then y	$\bar{x} + y$	1	1	0	1
Nand	$\overline{x \cdot y}$	1	1	1	0
Constant 1	1	1	1	1	1

Source: https://en.wikipedia.org/wiki/George_Boole, https://zhangruochi.com/Boolean-Functions-and-Gate-Logic/2019/05/20/

2) HISTORY OF COMPUTING

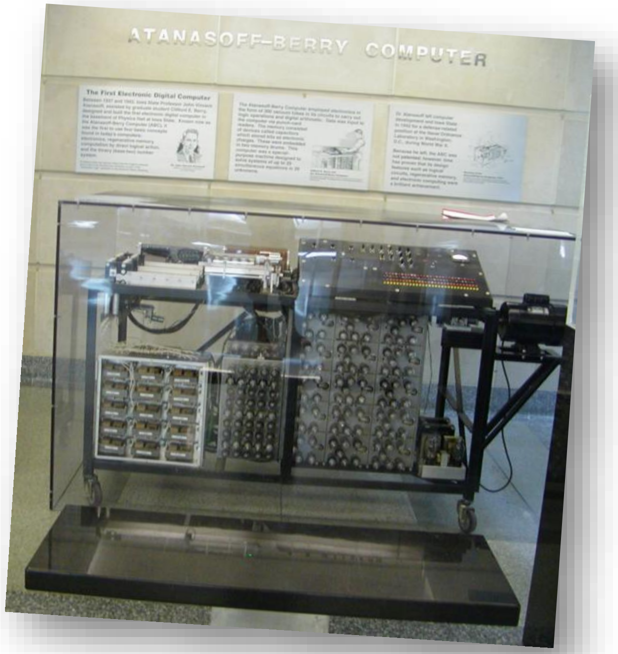
- In 1934, Johannes Von Neumann defines the computer architecture
- In 1936, Alan Türing define the notion of algorithm and of complexity
- In 1940, the Printed Circuit Board (PCB) is invented



Source: https://en.wikipedia.org/wiki/Printed_circuit_board

2) HISTORY OF COMPUTING

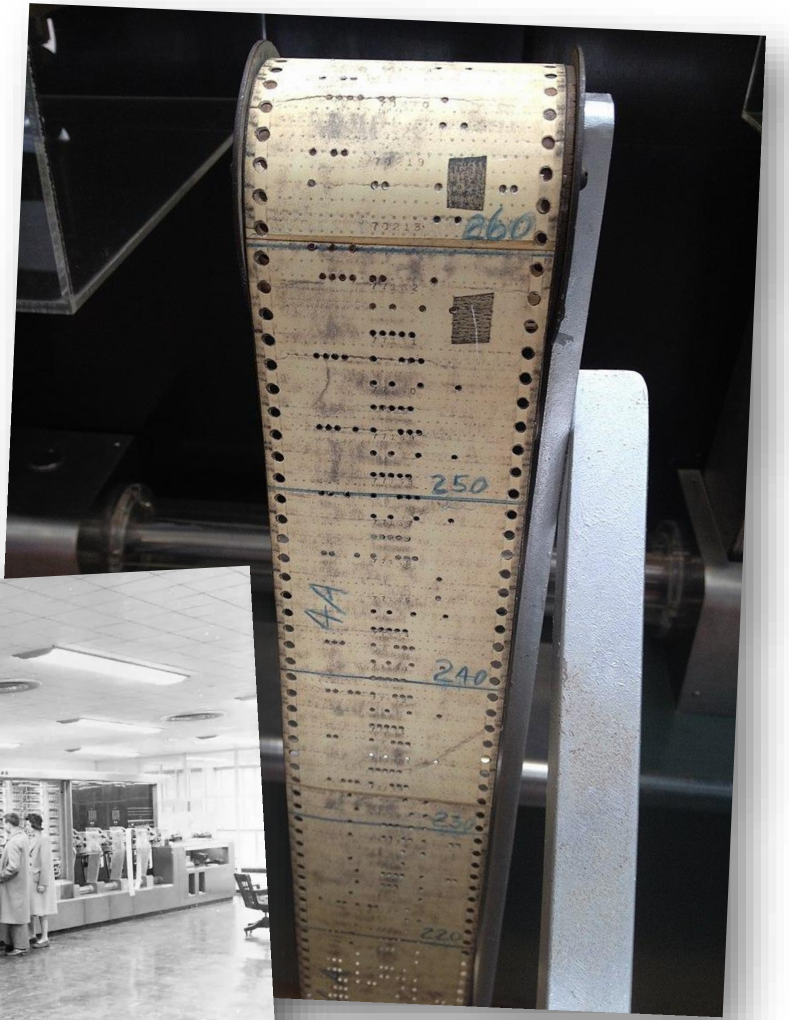
- In 1941, Konrad Zuse invents a motor-driven mechanical computer: Z1, then Z3. This last has 2 600 relays, its memory contained 64 numbers
- In 1942, the ABC, Atanasoff–Berry computer was created. Its frequency was 60Hz, it was able to compute 30 additions/subtractions per second. Its memory contained 60 numbers of 50 bits. It was the first computer to use binary



Source: https://en.wikipedia.org/wiki/Z1_(computer), https://en.wikipedia.org/wiki/Atanasoff%E2%80%93Berry_computer

2) HISTORY OF COMPUTING

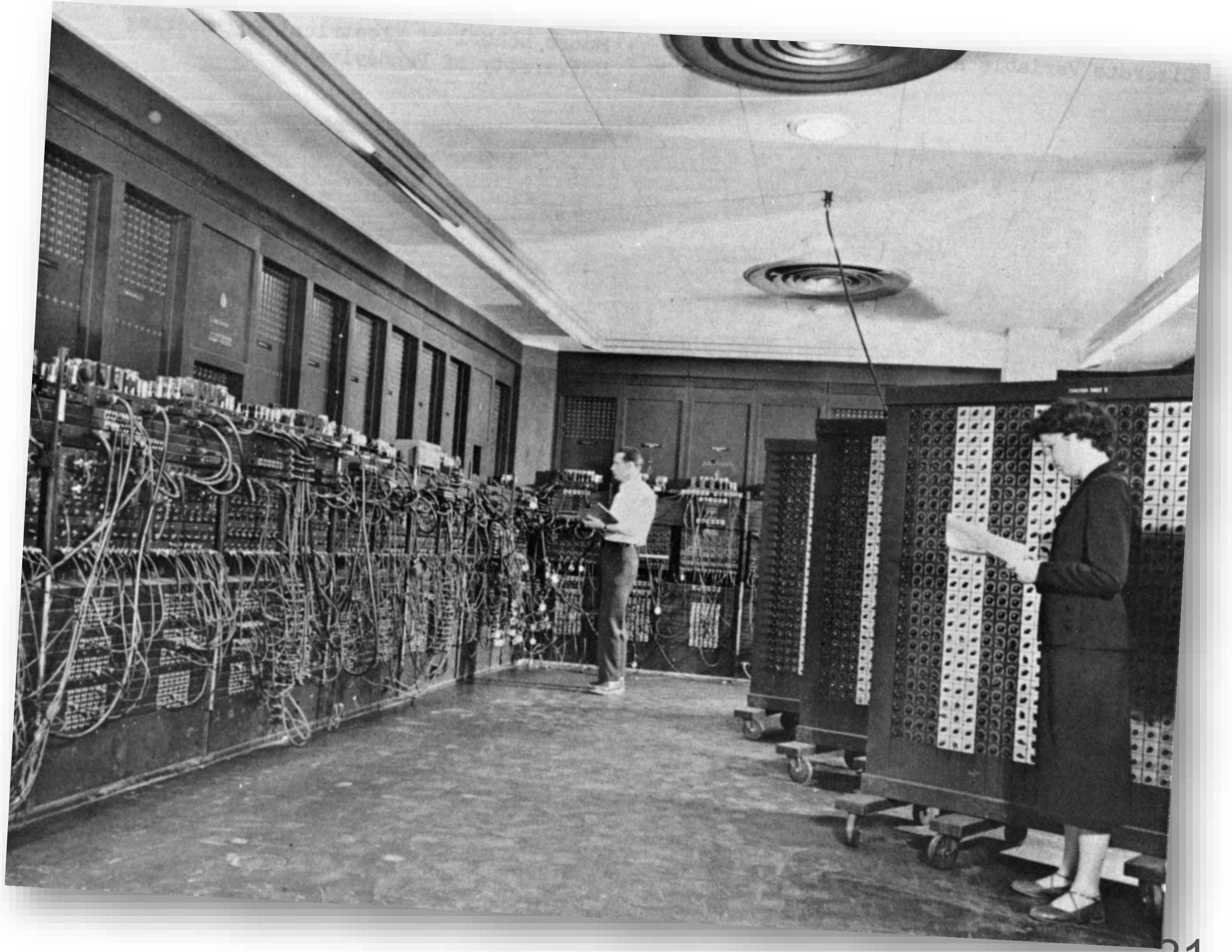
- In 1943, IBM creates the MARK I computer:
  - 17m long, 2.5m high, 5 tons, 800km of cables
- The Mark I had 60 sets of 24 switches for manual data entry and could store 72 numbers, each 23 decimal digits long. It could do 3 additions or subtractions in a second. A multiplication took 6 seconds, a division took 15.3 seconds, and a logarithm or a trigonometric function took over one minute.



Source: https://en.wikipedia.org/wiki/Harvard_Mark_I, https://www.britannica.com/technology/Harvard-Mark-I

2) HISTORY OF COMPUTING

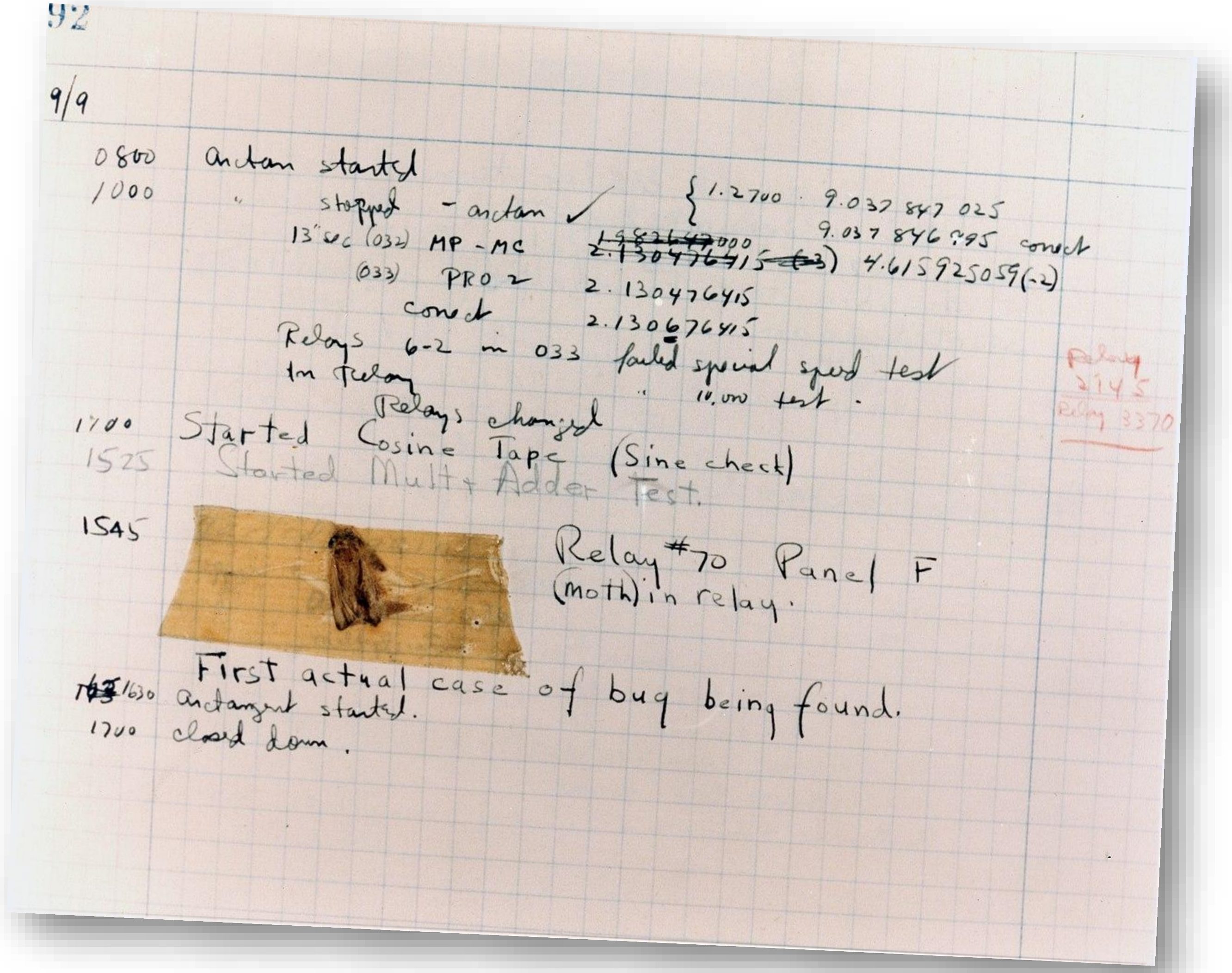
- In 1946, ENIAC (Electronic Numerical Integrator And Computer) in Philadelphia, Pennsylvania, USA: first computer without any mechanical part
- In 1956, ENIAC contained 18,000 vacuum tubes, 7,200 crystal diodes, 1,500 relays, 70,000 resistors, 10,000 capacitors, and approximately 5,000,000 hand-soldered joints. It weighed more than 30 short tons (27 t), was roughly 8 ft (2 m) tall, 3 ft (1 m) deep, and 100 ft (30 m) long, occupied 300 sq ft (28 m2) and consumed 150 kW of electricity



Source: https://en.wikipedia.org/wiki/ENIAC

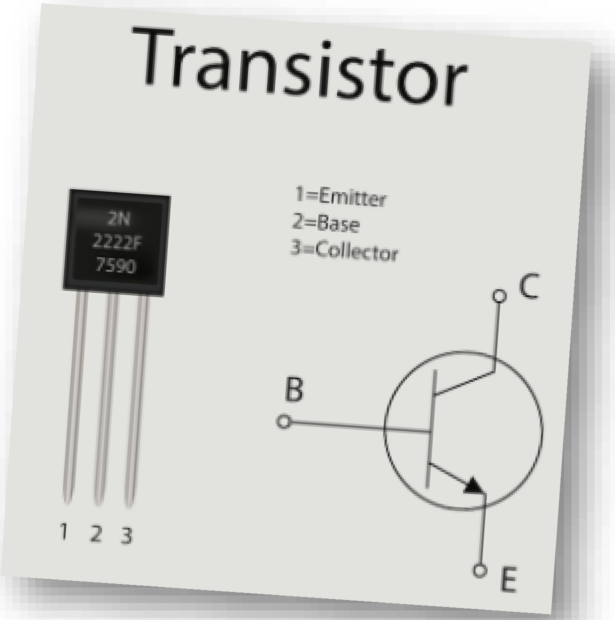
2) HISTORY OF COMPUTING

- The 9th September 1947 at 15:45, the first BUG in computer science was found!

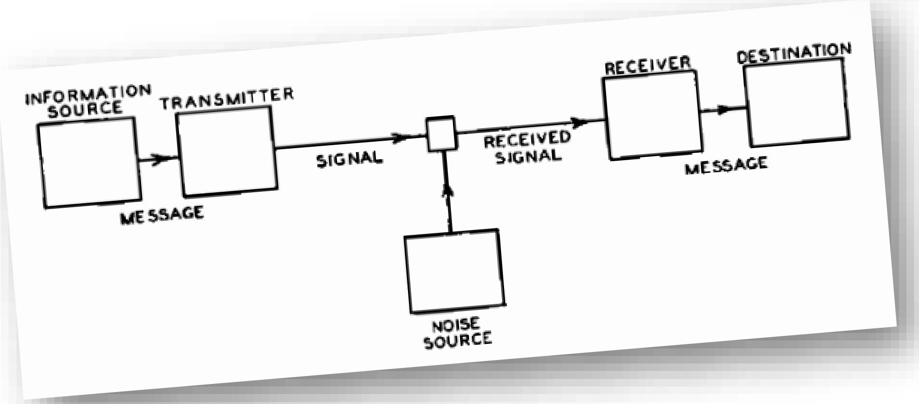


2) HISTORY OF COMPUTING

- In 1948, transistor is invented by John Bardeen, Walter Brattain and William Shockley. They received the Nobel prize in physics in 1956
- This invention reduces the size and the electric consumption of electronic circuits (and computers)
- In 1948, Claude Shannon defines the notion entropy in information theory:



$$H = -\sum_{i=1}^{N} p_i \log(p_i)$$



2) HISTORY OF COMPUTING

- In 1951, the UNIVAC computer is created
- The main memory consisted of tanks of liquid mercury implementing delay-line memory, arranged in 1,000 words of 12 alphanumeric characters each

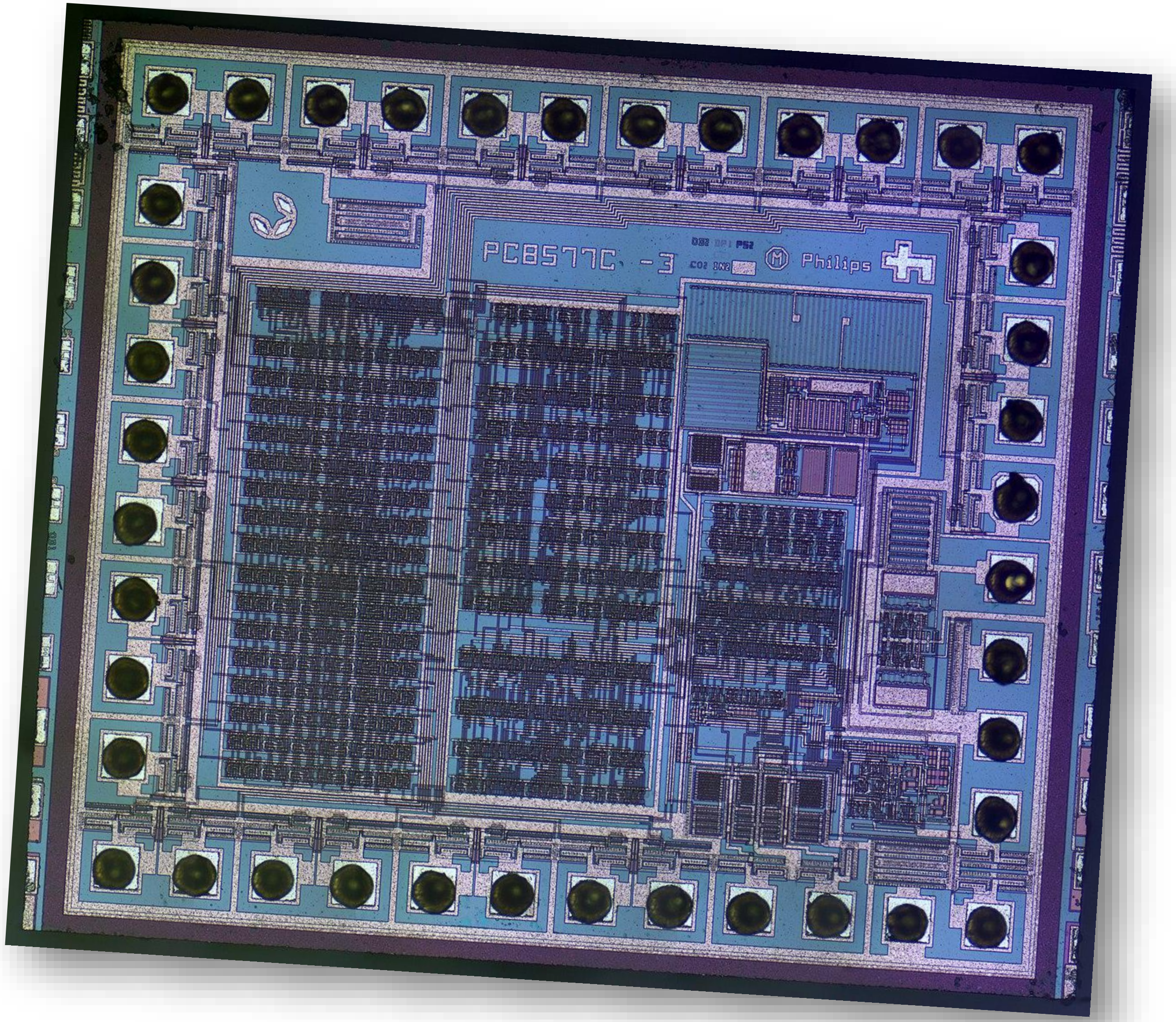
Operating systems [edit]

The 1107 was the first 36-bit, word-oriented machine with an architecture close to that which came to be known as that of the "1100 Series." It ran the EXEC I or EXEC II operating system, batch-oriented second-generation operating systems, typical of the early to mid-1960s. The 1108 ran EXEC II or EXEC 8. EXEC 8 allowed simultaneous handling of real-time applications, time-sharing, and background batch work. Transaction Interface Package (TIP), a transaction-processing environment, allowed programs to be written in COBOL whereas similar programs on competing systems were written in assembly language. On later systems, EXEC 8 was renamed OS 1100 and OS 2200, with modern descendants maintaining backwards compatibility.

Source: https://en.wikipedia.org/wiki/UNIVAC

2) HISTORY OF COMPUTING

- In 1958, the Integrated Circuit was invented by Texas Instrument
- It reduces the size and consumption of electronic devices



Source: https://en.wikipedia.org/wiki/Integrated_circuit

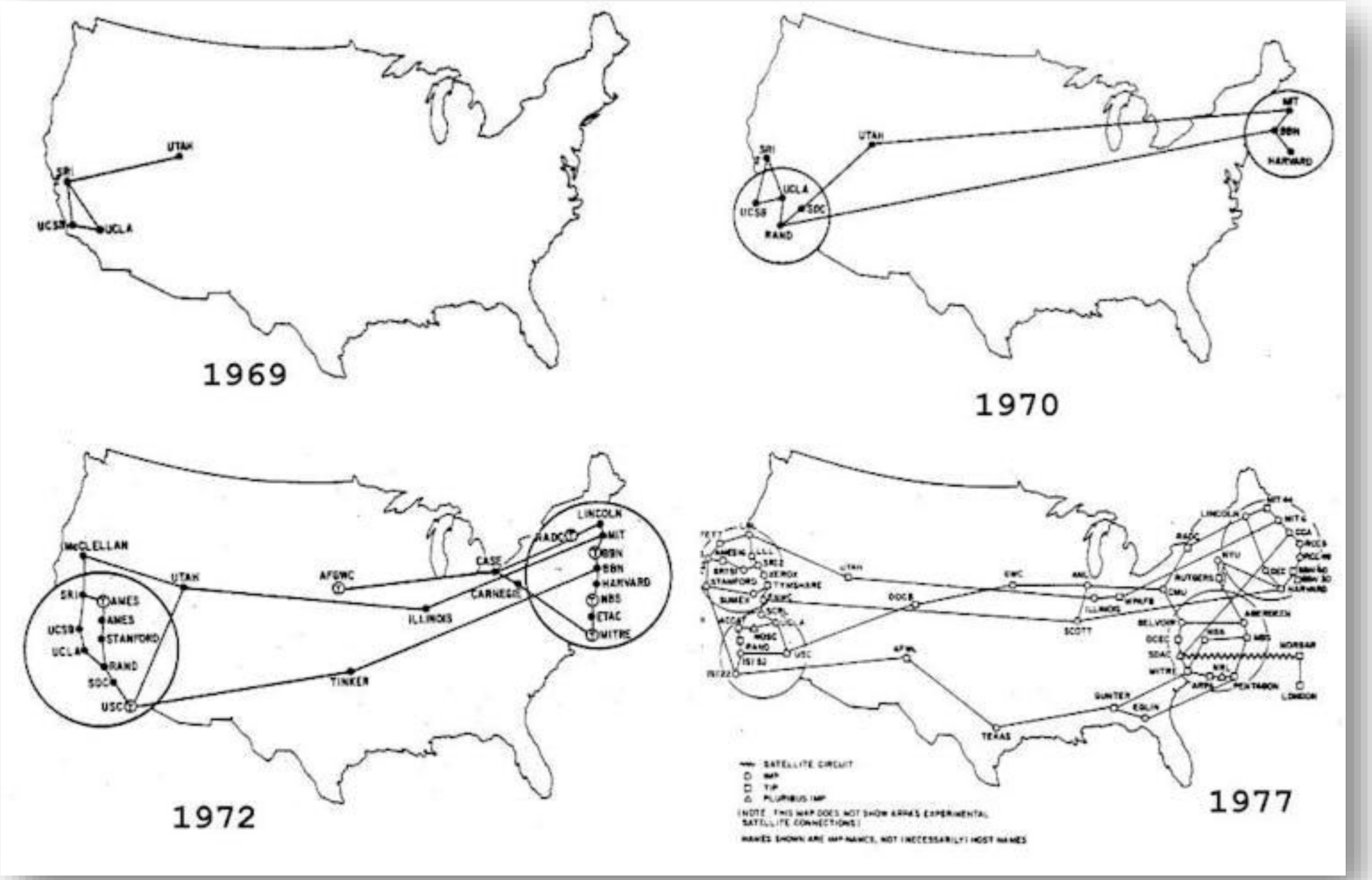
2) HISTORY OF COMPUTING

- In 1960, IBM 7000 is the first computer based on transistors
- In 1964, IBM 360 is created in 5 versions
- It is the ancestor of IBM AS/400 still in use in (several) companies

Source: https://en.wikipedia.org/wiki/IBM_AS/400

2) HISTORY OF COMPUTING

- In 1969, the Advanced Research Projects Agency Network (ARPANET) links 4 computers by a network across the USA:
  - University of California, Los Angeles (UCLA)
  - Stanford Research Institute (SRI)
  - University of California, Santa Barbara (UCSB)
  - University of Utah
- It is the ancestor of **INTERNET**



Source: https://images.theconversation.com/files/144160/original/image-20161102-27228-1iicfpi.jpg?ixlib=rb-4.1.0&q=45&auto=format&w=754&fit=clip

2) HISTORY OF COMPUTING

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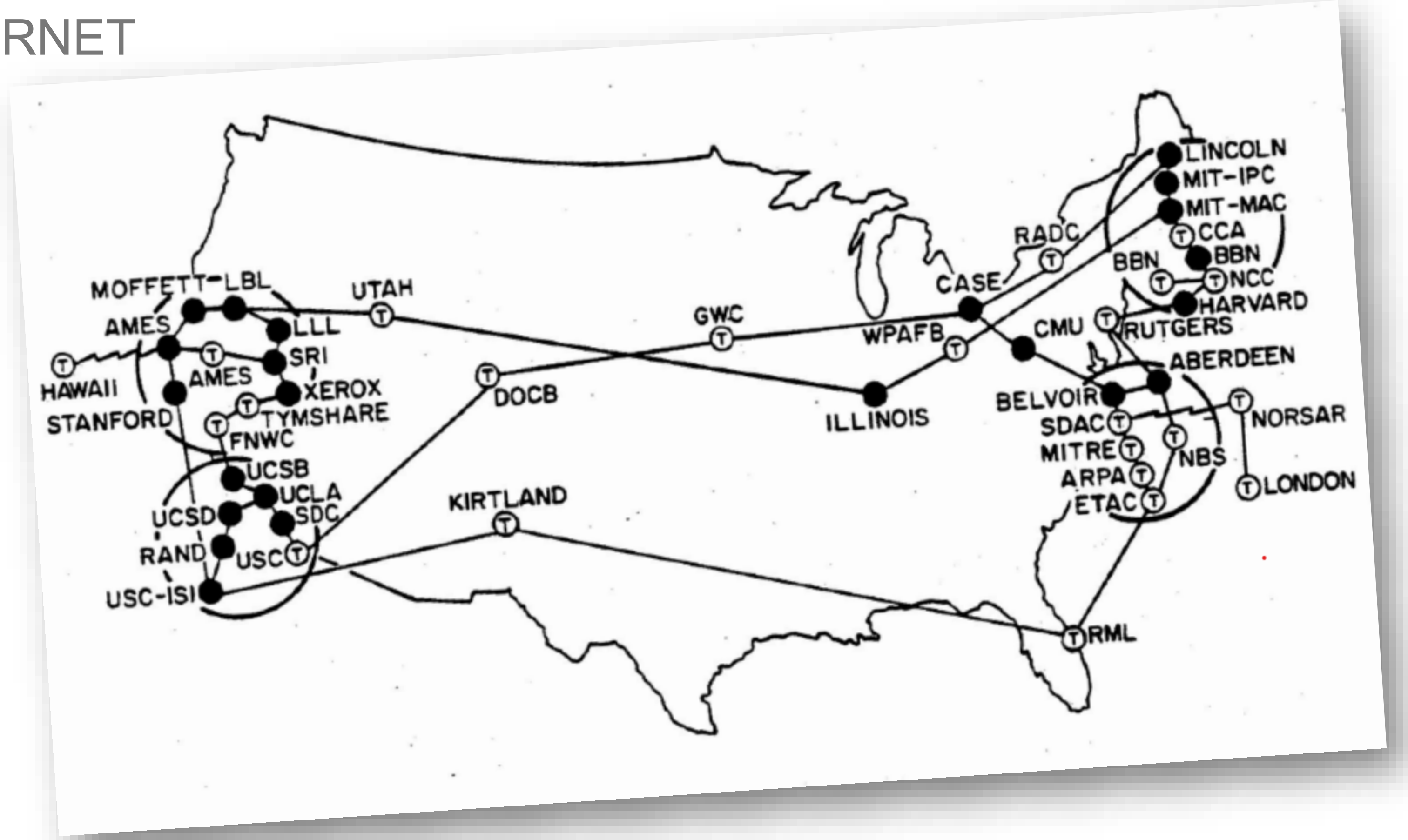
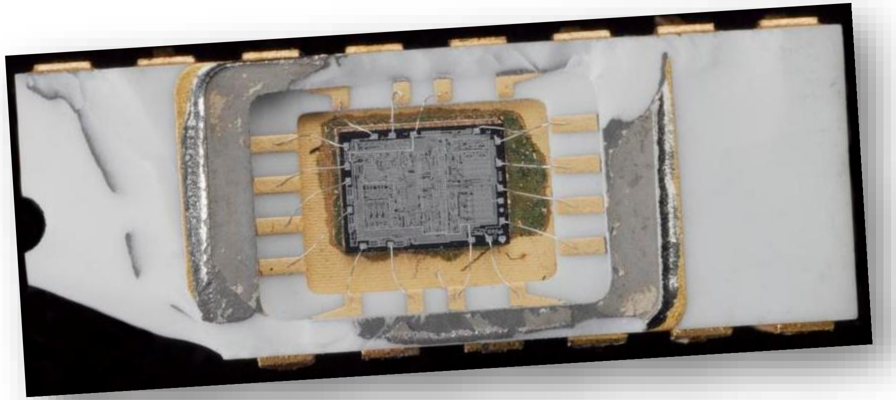


Source: https://en.wikipedia.org/wiki/ARPANET

2) HISTORY OF COMPUTING

- In 1971, the first microprocessor appears, it is the 4004 from Intel
  - 4-bits operations, frequency 108kHz, 640 bytes of memory, 60 000 operations/second



- In 1972, Intel creates the 8008
  - 8-bits operations, frequency 200kHz, 16 KB of memory

- The floppy disk also appears in 1972

Source: https://en.wikipedia.org/wiki/Intel_4004, https://en.wikipedia.org//wiki/Floppy_disk

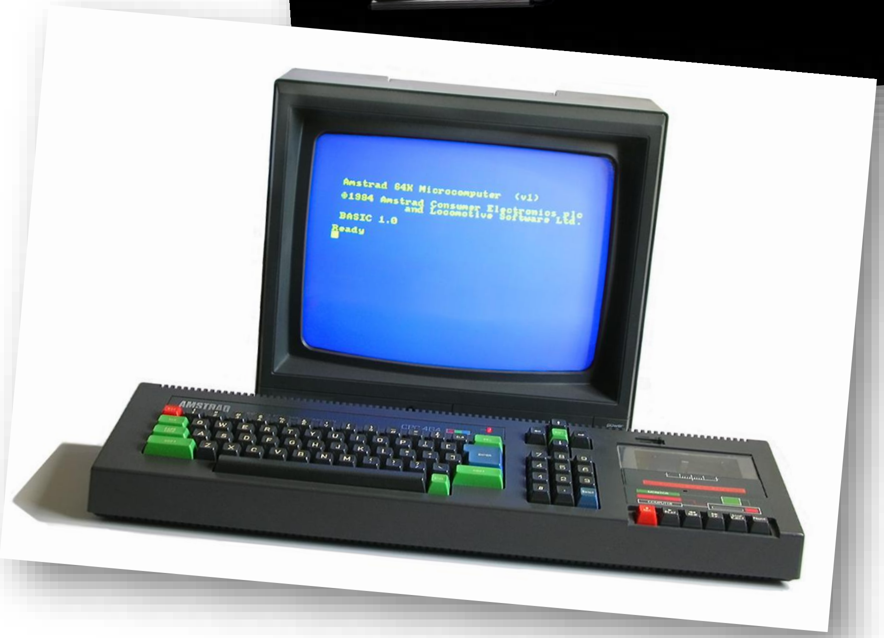
2) HISTORY OF COMPUTING

- In 1973, the first microcomputer is french, it is the R2E Micral based on an Intel 8080 processor
- In 1973, IBM launches the Winchester hard disk drive: 35 MB and 70 MB
- In 1976, the first Apple Computer is created. In 1977, the Apple II from Apple Computer is a commercial success

Source: https://en.wikipedia.org/wiki/Micral, https://en.wikipedia.org/wiki/Hard_disk_drive, https://en.wikipedia.org/wiki/History_of_hard_disk_drives, https://en.wikipedia.org/wiki/Apple_Inc.

2) HISTORY OF COMPUTING

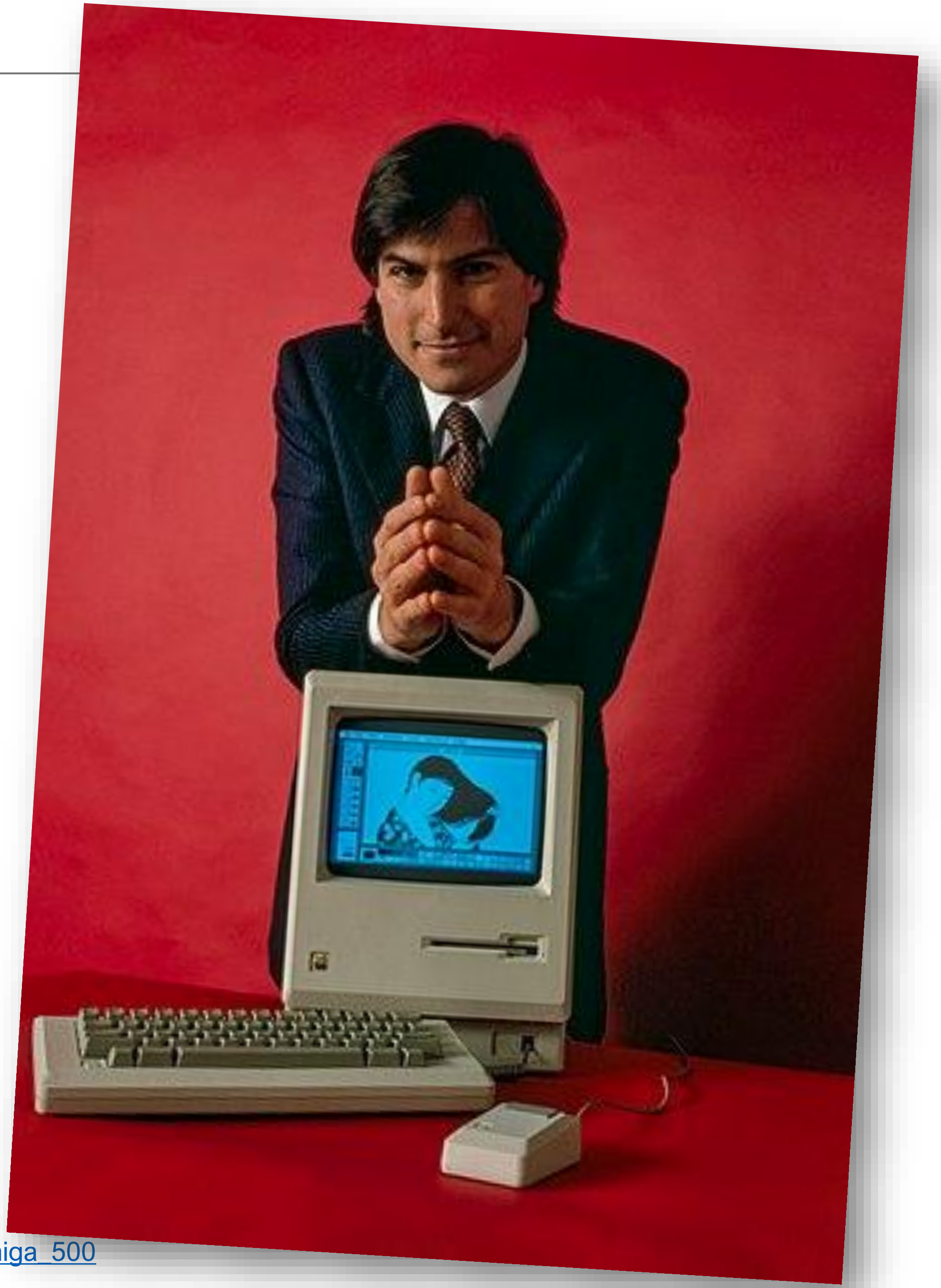
- Then came Commodore PET, Tandy TRS-80, ZX81, Commodore 64, Oric 1
- In 1981, IBM creates the IBM PC, the ancestor of all PC computers
- In 1983, Thomson creates the MO5 followed by the TO7
- In 1985, Amstrad sells CPC 464 and CPC 6128



Source: https://en.wikipedia.org/wiki/Thomson_TO7, https://en.wikipedia.org/wiki/Amstrad_CPC

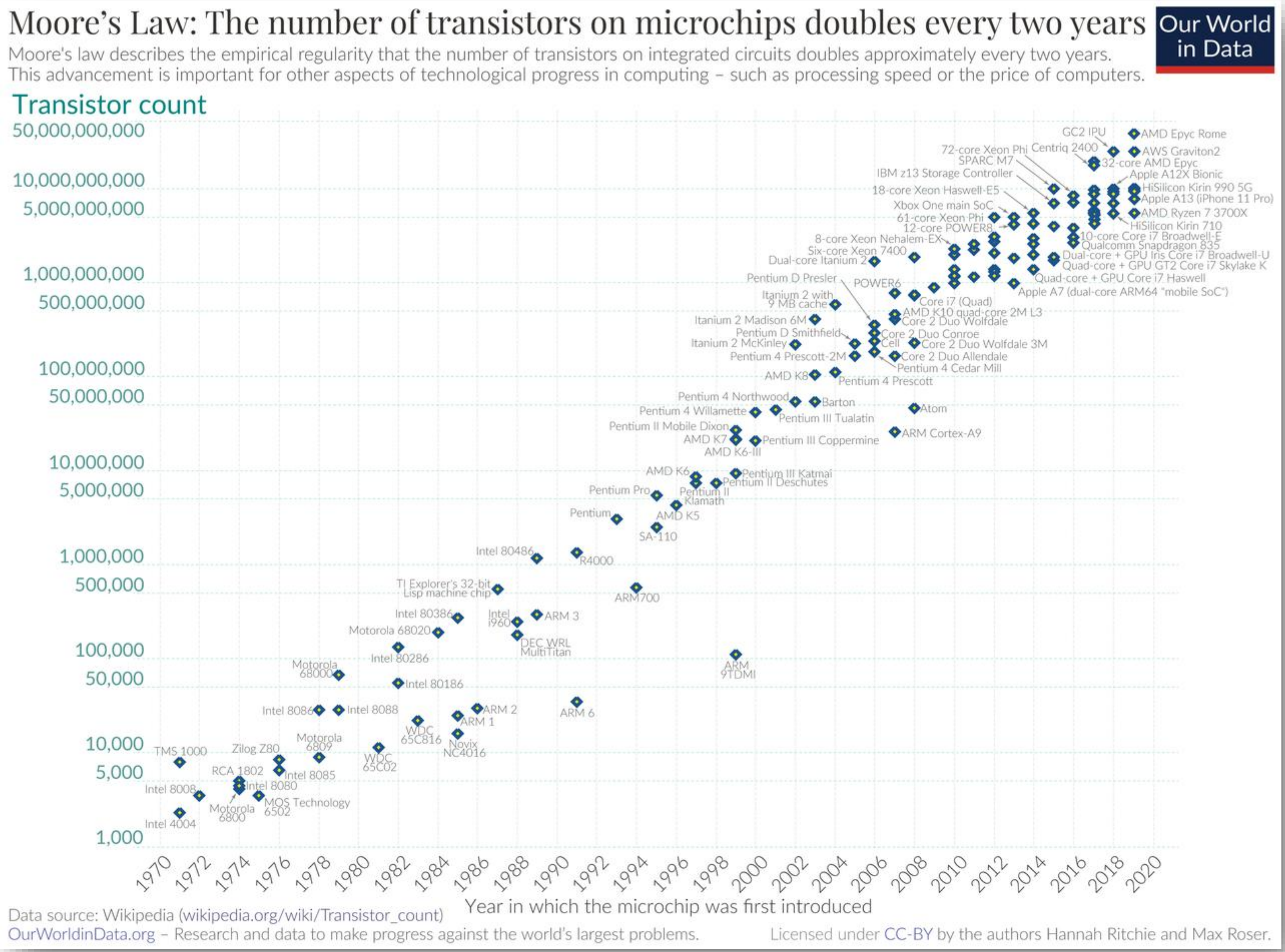
2) HISTORY OF COMPUTING

- In 1984, the Apple Macintosh is created
- In 1986, 8-bits computers are replaced by 16-bits computers, for instance Atari 520 ST & Amiga 500



Source: https://en.wikipedia.org/wiki/Mac_(computer), https://en.wikipedia.org/wiki/Atari_ST, https://en.wikipedia.org/wiki/Amiga_500

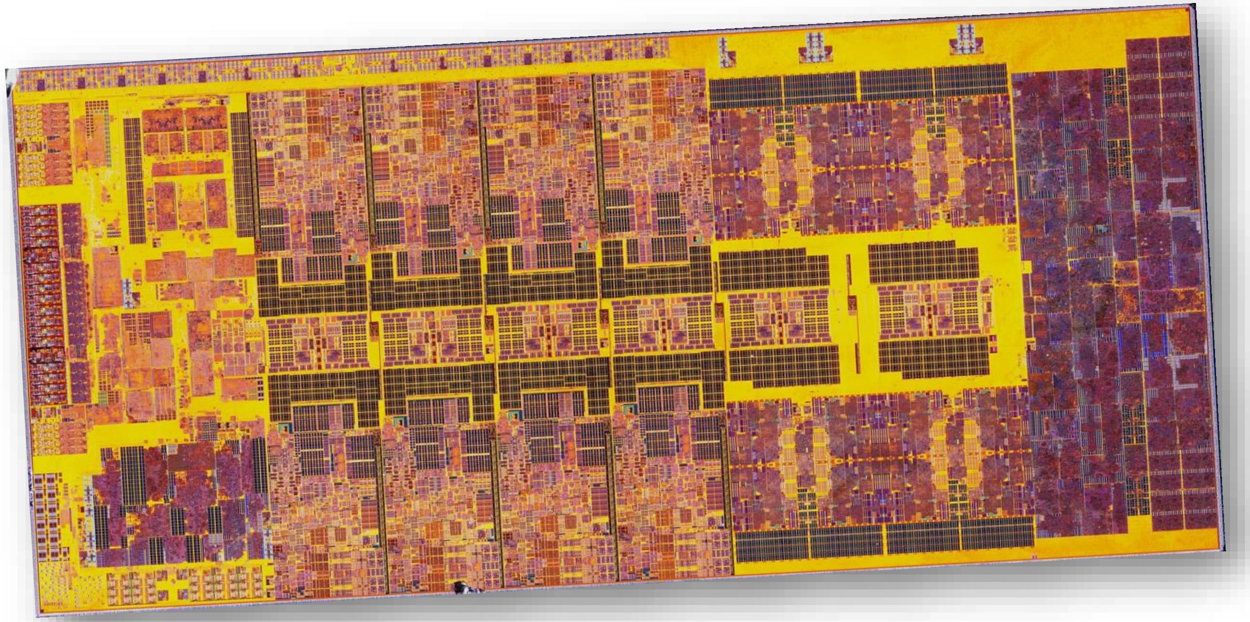
- Internet is developed all around the World
- Microsoft creates Windows 3.11, Windows 95, 98, Me, NT, 2000, XP, 7, 10…
- The computer becomes a standard product
- The microprocessors follow Moore's law: the number of transistors doubles every 2 years
- In 2019, AMD Ryzen 9 3900X has up to 9.89 billion transistors



Source: https://devicetests.com/how-many-transistors-are-in-a-cpu, https://en.wikipedia.org/wiki/Moore's_law

2) HISTORY OF COMPUTING

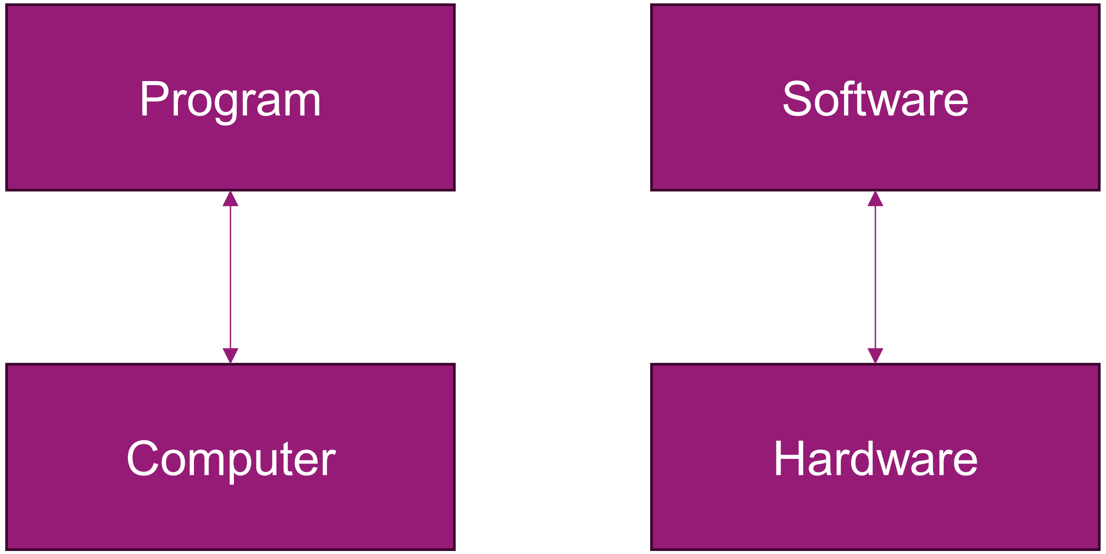
- Recently, 13th generation of Intel 64-bits processors: Raptor Lake family
  - CPU:
    - Cores, Up to 8 Raptor Cove performance cores (P-core), Up to 16 Gracemont efficiency cores (E-core)
    - L2 cache for the P-core increased to 2 MB and for the E-core cluster to 4 MB, Up to 36 MB L3 cache
  - GPU: Up to 96 Execution Units, Intel Iris Xe-LP microarchitecture, Up to 1.65 GHz frequency
  - I/O: Up to DDR5-5600, Up to 28 PCI Express lanes, Up to 5 USB 3.2 20 Gbit/s ports



Source: https://en.wikipedia.org/wiki/Raptor_Lake

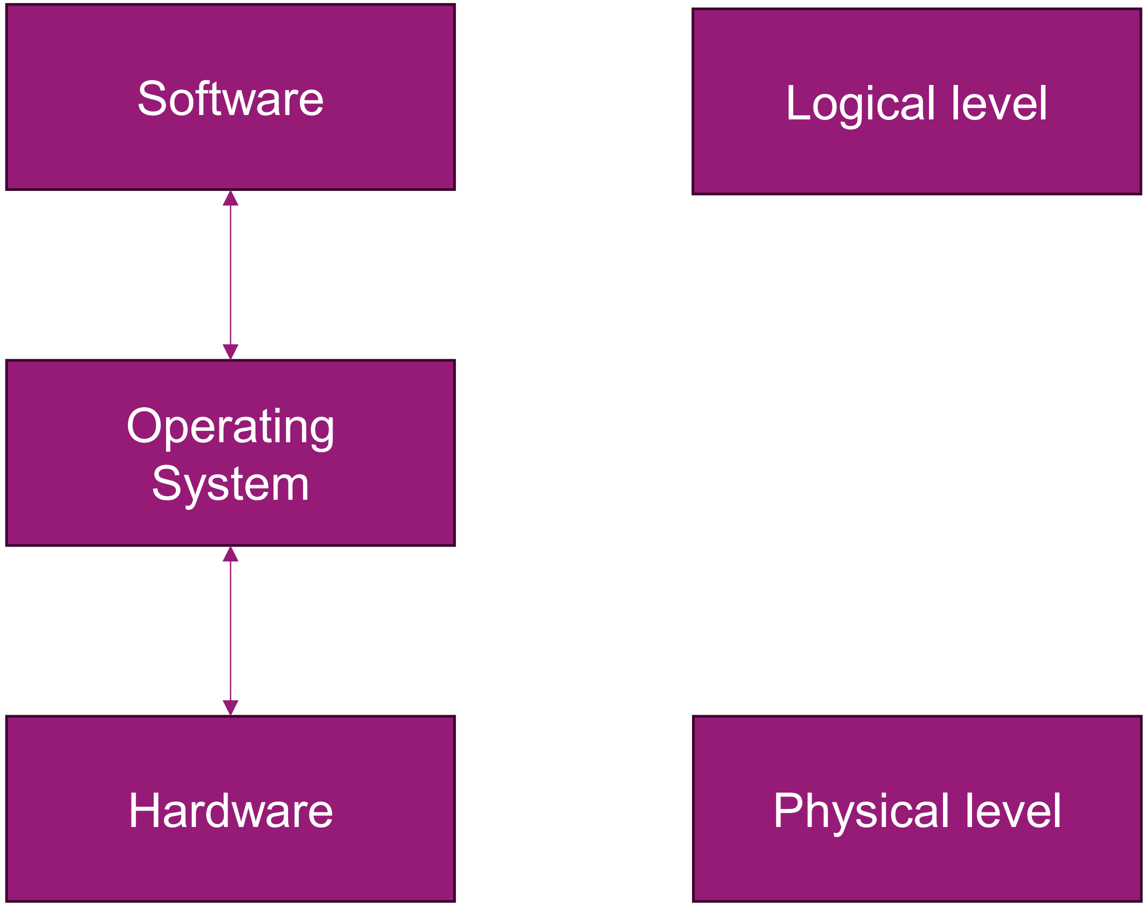
3) OPERATING SYSTEMS

- On the first computers:

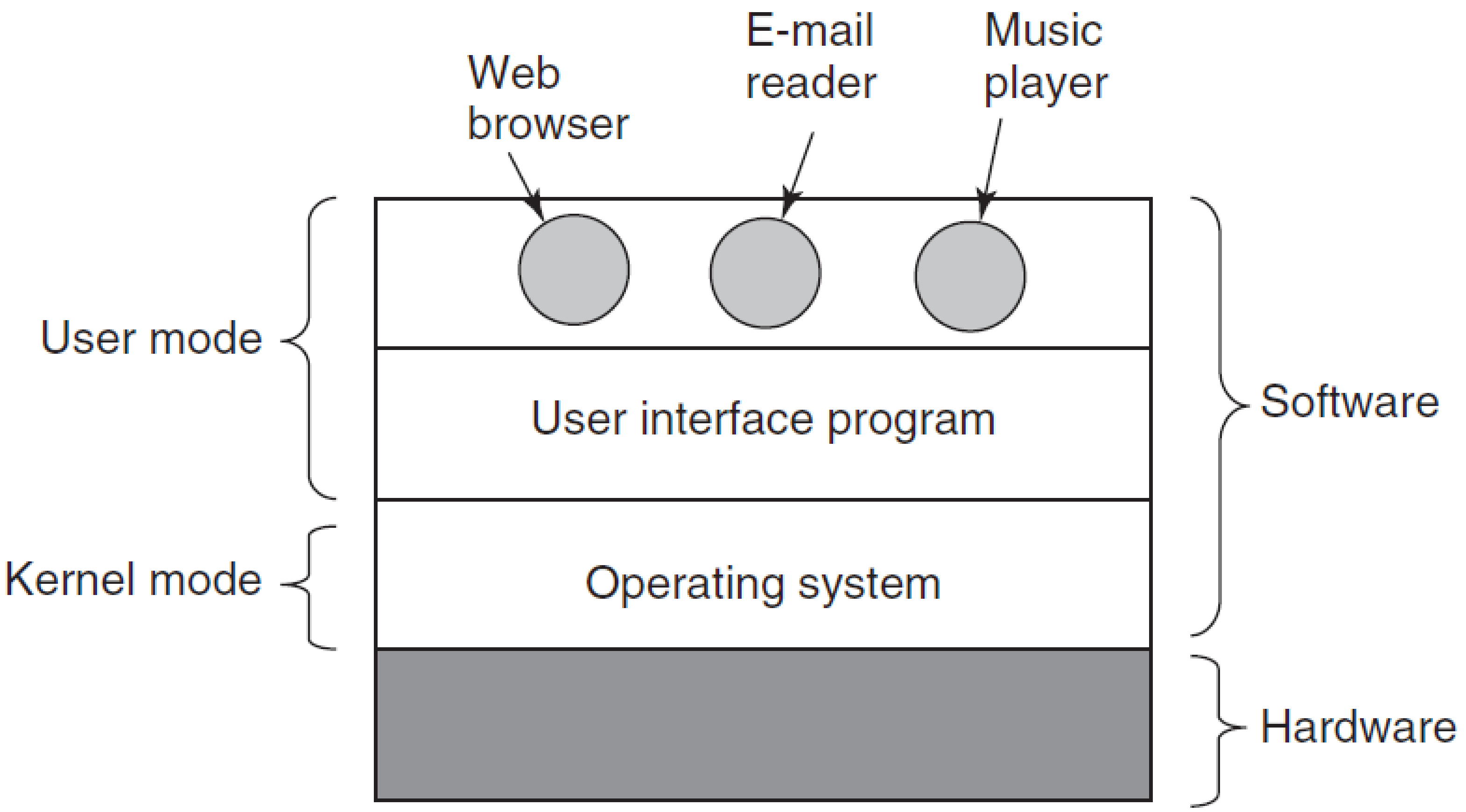


3) OPERATING SYSTEMS

- On actual computers:



3) OPERATING SYSTEMS



Source: Book Modern Operating Systems.

ANY QUESTIONS?